Rishi Malhan

Ph.D. Robotics and AI

University of Southern California

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Github: https://www.github.com/rishimalhan

Portfolio: https://www.rmalhan.com

Linkedin: https://www.linkedin.com/in/rishi-malhan/

Roboticist with expertise in leveraging artificial intelligence for manipulation systems with a focus on task and motion planning, closed loop learning, and human-robot interaction projects. Self-driven and customer focused individual with a goal to develop impactful, and scalable algorithmic solutions to enhance business performance and reduce costs

RESEARCH INTEREST

Robotics, Physics-informed Artificial Intelligence, Advanced Manufacturing, Task and Motion Planning, Multi-Robot Systems, and Large language models

TECHNICAL SKILLS

Programming & Robots: Python, C++, Matlab, Kuka Sunrise, Epson RC+, ABB RobotStudio, Fanuc Roboguide
Tools and Libraries: Robot Operating System (ROS), MoveIt, OpenRAVE, Vrep, prpy, PyTorch, Rviz, FCL, kdl, NLOpt, OpenCV, open3d, scikit-learn, pandas, Eigen, libnabo, MeshLab, asyncIO, tornado (coroutines, ioloop), threading (locks, events, conditions), Docker, CMake, git, kibana, mongoDB, faster-whisper, tortoise, scikit-LLM, Huggingface, Runpod

EDUCATION

Ph. D. Robotics and AI, *University of Southern California*, *Advisor: Dr. S. K. Gupta* **May 2018 - May 2022**

- **Dissertation:** Robot trajectory generation and placement under motion constraints **Coursework:** Numerical optimization, Artificial intelligence, Robotics, Machine learning, Deep learning and computational physics

M.S. Mechanical Engineering, *University of Southern California, USA* **August 2016 - May 2018**

- **Honors:** Academic excellence award, Aerospace and Mechanical Engineering

B.TECH Mechanical Engineering, *Nirma University, India* **August 2012 - May 2016**

- **Honors:** Gold medals for academic excellence, Year 1 - 4

EXPERIENCE

Berkshire Grey Inc. *Senior Roboticist* **June 2022 - Present**

- Confidential robotics system (a) Lead the development of multi-arm coordination algorithm based on spatial reservation (b) Developed grasp affordances and task planning framework for a novel package handling end-effector (c) Optimized robot position with an objective to improve reachability (d) Deep dived and reduced planning failures by 60% and torque faults by 20% (e) Worked on integration of force-torque sensor for grasp execution
- Defined KPIs from statement of work and built metrics dashboard to characterize robot performance
- Wrote production code optimized for efficiency using multi-threading and async programming with exception handling
- Spearheaded customer facing demos with simple yet effective presentation strategies

Berkshire Grey Inc. *Ph.D. Intern* **May - August, 2020 & 2021**

- Improved Singulation speeds by 3-5% (significant at scale) (a) Developed classification algorithms to root-cause SKU drops on production data (b) Formulated acceleration constraints for time optimal robot path parameterization

University of Southern California *Graduate Research Assistant* **January 2017 - May 2022**

- Design and implementation of Task and motion planning algorithms with impedance control for assembly of a small-medium sized satellite (CUBE-SAT)
- Multi-arm manipulation of deformable and viscoelastic composite sheets for automation of hand-layup process in collaboration with Aerospace industry
- Realized a multi-robot grasp planner to manipulate composite sheets with Gaussian Process Regression based online plan refinement
- Developed heuristics for depth-first BnB algorithm to find optimal sequence of mobile manipulator base placements
- Supportless additive manufacturing of complex geometry shell components by using higher degrees of freedom from a multi-arm setup
- Developed graph-based search for planning robot motion over semi-constrained tool paths
- Coverage planning under constraints for fast and efficient 3D reconstruction of objects using depth camera

SIDE PROJECTS

- Prompt engineer LLMs to bridge the gap between high-level natural language and low-level functions
- Characterize the uncertainty and evaluate inference capabilities of LLMs applied towards human-robot interaction

AWARDS/HONORS

- Best Paper Award (Second Place) in 2021 ASME Manufacturing Science and Engineering Conference for the paper "P. M. Bhatt, R. K. Malhan, P. Rajendran, A. V. Shembekar, and S. K. Gupta. Trajectory-dependent compensation scheme to reduce manipulator execution errors for manufacturing applications. ASME Manufacturing Science and Engineering Conference, June 2021"
- First Place in 2019 SME Aerospace and Defense Manufacturing Conference Poster Challenge for poster titled "Robotic Assistants for Surface Finishing" by A.M. Kabir, R.K. Malhan, A.V. Shembekar, B.C. Shah, and S.K. Gupta
- Third Place in 2019 SME Aerospace and Defense Manufacturing Conference Poster Challenge for poster titled "Realizing Supportless Extrusion-Based Additive Manufacturing through Use of Robots" by P.M. Bhatt, R.K. Malhan, and S.K. Gupta
- Best Paper Award (Third Place) in 2018 ASME Manufacturing Science and Engineering Conference for the paper "A.M. Kabir, A.V. Shembekar, R.K. Malhan, R.S. Aggarwal, J.D. Langsfeld, B.C. Shah, and S.K. Gupta. Robotic finishing of interior regions of geometrically complex parts. ASME Manufacturing Science and Engineering Conference, College Station, TX, June 2018"
- Judge's Choice Award at 2018 Reusable Abstractions of Manufacturing Processes Workshop for poster titled "Hybrid Cell for Multi-Layer Prepreg Composite Sheet Layup" by R. Malhan, A. Kabir, B. Shah, T. Centea and S.K. Gupta"

PATENT

Malhan, Rishi, et al. "Hybrid formation of multi-layer prepreg composite sheet layup." U.S. Patent No. 11628624

Malhan, Rishi, et al. "Hybrid formation of multi-layer prepreg composite sheet layup." U.S. Patent No. 11919239

FELLOWSHIPS

Ph.D. Fellowship Award 2018-2019, Viterbi School of Engineering, University of Southern California

SERVICE

- Associate Editor, International Conference on Robotics and Automation (ICRA 2023)
- Reviewer for Conferences: International Conference on Robotics and Automation, Intelligent Robots and Systems, Conference on Automation Science and Engineering, Manufacturing Science and Engineering Conference, and ASME Design Engineering Technical Conferences & Computers and Information in Engineering Conference
- Reviewer for Journals: Transactions on Automation Science and Engineering, Robotics and Computer Integrated Manufacturing, Additive Manufacturing, and Robotics and Automation Letters

JOURNAL PUBLICATIONS

[1] R.K. Malhan and S.K. Gupta. The Role of Deep Learning in Manufacturing Applications: Challenges and Opportunities. ASME Journal of Computing and Information Science in Engineering, 23(6): 060816, December 2023.

[2] R.K. Malhan and S.K. Gupta. Planning algorithms for acquiring high fidelity pointclouds using a robot for accurate and fast 3D reconstruction. Robotics and Computer-Integrated Manufacturing (RCIM). 2022.

[3] R.K. Malhan, S. Thakar, A.M. Kabir, P. Rajendran, P.M. Bhatt, and S.K. Gupta. Generation of configuration space trajectories over semi-constrained cartesian paths for robotic manipulators. IEEE Transactions on Automation Science and Engineering. (TASE) 2022.

[4] R.K. Malhan, R.J. Joseph, P.M. Bhatt, B. Shah, and S.K. Gupta. Algorithms for improving speed and accuracy of automated 3D reconstruction with a depth camera mounted on an industrial robot. ASME Journal of Computing and Information Science in Engineering, 22(3): 031012, June 2022.

[5] S. Thakar, S. Srinivasan, S. Al-Hussaini, P.M. Bhatt, P. Rajendran, Y. J. Yoon, N. Dhanaraj, R.K. Malhan, M. Schmid, V.N. Krovi and S.K. Gupta. A survey of wheeled mobile manipulation: A decision making perspective. ASME Journal of Mechanisms and Robotics. 2022.

[6] S Thakar, RK Malhan, PM Bhatt, and S.K. Gupta. Area-coverage planning for spray-based surface disinfection with a mobile manipulator. Robotics and Autonomous Systems, 147: 103920, January 2022.

[7] P.M. Bhatt, A. Kulkarni, A. Kanyuck, R. K. Malhan, L. S. Santos, S. Thakar, H. A. Bruck, and S.K. Gupta. Automated process planning for conformal wire arc additive manufacturing. International Journal of Advanced Manufacturing Technology. 2022

[8] P.M. Bhatt, A. Kulkarni, R.K. Malhan, B.C. Shah, Y.J. Yoon, and S.K. Gupta. Automated planning for robotic multi-resolution additive manufacturing. ASME Journal of Computing and Information Science in Engineering, 22(2): 021006, April 2022.

[9] Y.W. Chen, R.J. Joseph, A. Kanyuck, S. Khan, R.K. Malhan, O.M. Manyar, Z. McNulty, B. Wang, J. Barbič, S.K. Gupta. A digital twin for automated layup of prepreg composite sheets. ASME Journal of Manufacturing Science and Engineering, 144(4):041010, April 2022.

[10] A. M. Kabir, S. Thakar, R. K. Malhan, A. V. Shembekar, B. C. Shah, and S. K. Gupta. Generation of Synchronized Configuration Space Trajectories with Workspace Path Constraints for an Ensemble of Robots. The International Journal of Robotics Research (IJRR). 2021.

[11] P.M. Bhatt, R.K. Malhan, P. Rajendran, B.C. Shah, S. Thakar, Y.J. Yoon, and S.K. Gupta. Image-based surface defect detection using deep learning: A review. ASME Journal of Computing and Information Science in Engineering, 21(4):040801 August 2021.

[12] R. K. Malhan, A. V. Shembekar, A. M. Kabir, P. M. Bhatt, B. Shah, S. Zanio, S. Nutt, and S. K. Gupta. Automated Planning for Robotic Layup of Composite Prepreg. Robotics and Computer-Integrated Manufacturing. 2020.

[13] P. M. Bhatt, R. K. Malhan, P. Rajendran, B. C. Shah, S. Thakar, Y. J. Yoon, and S. K. Gupta. Image-based Surface Defect Detection Using Deep Learning: A Review. Journal of Computing and Information Science in Engineering. 2020.

[14] P. M. Bhatt, R. K. Malhan, A. V. Shembekar, Y. J. Yoon, and S.K. Gupta. Expanding capabilities of additive manufacturing through use of robotics: A survey. Additive Manufacturing. 31:100933, 2020.

[15] P. M. Bhatt, R. K. Malhan, P. Rajendran, and S.K. Gupta. Building free-form thin shell parts using supportless extrusion-based additive manufacturing. Additive Manufacturing. 101003, 2020.

CONFERENCE PUBLICATIONS

[1] R.K. Malhan and S.K. Gupta. Finding optimal sequence of mobile manipulator placements for automated coverage planning of large complex parts. ASME Computers and Information in Engineering Conference, St. Louis, MO, August 2022.

[2] N. Dhanaraj, R. Malhan, H. Nemlekar, S. Nikolaidis, and S.K. Gupta. Human-Guided Goal Assignment to Effectively Manage Workload for a Smart Robotic Assistant. IEEE International Conference on Robot & Human Interactive Communication, Naples, Italy, August 2022.

[3] R. Malhan, R. J. Joseph, P. M. Bhatt, B. Shah, and S. K. Gupta. Fast, Accurate, and Automated 3D Reconstruction Using a Depth Camera Mounted on an Industrial Robot. ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, Online, Virtual, August 2021.

[4] N. Dhanaraj, Y.J. Yoon, R. Malhan, P.M. Bhatt, S. Thakar and S.K. Gupta. A mobile manipulator system for accurate and efficient spraying on large surfaces. 3rd International Conference on Industry 4.0 and Smart Manufacturing, Linz, Austria, November 2021.

[5] P.M. Bhatt, A. Kulkarni, R.K. Malhan, and S.K. Gupta. Optimizing Part placement for improving accuracy of robot-based additive manufacturing. IEEE International Conference on Robotics and Automation, Xi’an, China, May 2021.

[6] O. M. Manyar, J. A. Desai, N. Deogaonkar, R. J. Joseph, R. K. Malhan, Z. McNulty, B. Wang, J. Barbič, S. K. Gupta. A simulation-based grasp planner for enabling robotic grasping during composite sheet layup. IEEE International Conference on Robotics and Automation, Xi'an, China, May 2021.

[7] P. M. Bhatt, R. K. Malhan, P. Rajendran, A. V. Shembekar, and S. K. Gupta. Trajectory dependent compensation scheme to reduce manipulator execution errors for manufacturing applications. ASME Manufacturing Science and Engineering Conference, June 2021.

[8] R.K. Malhan, R. Jomy Joseph, A.V. Shembekar, A.M. Kabir, P.M. Bhatt, and S.K. Gupta. Online Grasp Plan Refinement for Reducing Defects During Robotic Layup of Composite Prepreg Sheets. IEEE International Conference on Robotics and Automation (ICRA), Paris, France, May 31-June 4, 2020.

[9] A.M. Kabir, S. Thakar, P.M. Bhatt, R.K. Malhan, P. Rajendran, B.C. Shah, and S.K. Gupta. Incorporating motion planning feasibility considerations during task-agent assignment to perform complex tasks using mobile-manipulators. IEEE International Conference on Robotics and Automation, Paris, France, June 2020.

[10] P. M. Bhatt, C. Gong, A. M. Kabir, R. K. Malhan, B. C. Shah, and S. K. Gupta. Incorporating Tool Contact Considerations in Tool-Path Planning for Robotic Operations. ASME Manufacturing Science and Engineering Conference, Cincinnati, OH, June 2020.

[11] R.K. Malhan, A.V. Shembekar, R.J. Joseph, A.M. Kabir, P.M. Bhatt, B.C. Shah, W. Zhao, S. Nutt, and S.K. Gupta. A smart robotic cell for automating composite prepreg layup. North 19 America Society for the Advancement of Material and Process Engineering (SAMPE), Seattle, WA, USA, May 2020.

[12] R.K. Malhan, A.M. Kabir, B.C. Shah, and S.K. Gupta. Identifying Feasible Workpiece Placement with Respect to Redundant Manipulator for Complex Manufacturing Tasks. IEEE International Conference on Robotics and Automation (ICRA 2019), Montreal, Canada, May 20 - May 24, 2019.

[13] P.M. Bhatt, A.M. Kabir, R.K. Malhan, B.C. Shah, A.V. Shembekar, Y.J. Yoon, and S.K. Gupta. A Robotic Cell for Multi-Resolution Additive Manufacturing. IEEE International Conference on Robotics and Automation (ICRA 2019), Montreal, Canada, May 20 - May 24, 2019.

[14] A.M. Kabir, A. Kanyuck, R.K. Malhan, A.V. Shembekar, S. Thakar, B.C. Shah, and S.K. Gupta. Generation of Synchronized Configuration Space Trajectories of Multi-Robot Systems. IEEE International Conference on Robotics and Automation (ICRA), Montreal, Canada May 20 - May 24, 2019.

[15] R.K. Malhan, A.M. Kabir, B. Shah, T. Centea, and S.K. Gupta. Determining Feasible Robot Placements in Robotic Cells for Composite Prepreg Sheet Layup. ASME Manufacturing Science and Engineering Conference, Erie, PA, June 2019.

[16] P.M. Bhatt, R.K. Malhan, and S.K. Gupta. Computational foundations for using three degrees of freedom build platforms to enable supportless extrusion-based additive manufacturing. ASME Manufacturing Science and Engineering Conference, Erie, PA, June 2019.

[17] P.M. Bhatt, A.M. Kabir, R.K. Malhan, A.V. Shembekar, B.C. Shah, and S.K. Gupta. Concurrent Design of Tool-Paths and Impedance Controllers for Performing Area Coverage Operations in Manufacturing Applications

under Uncertainty. IEEE International Conference on Automation Science and Engineering (CASE), Vancouver, Canada, August 2019.

[18] S. Thakar, A. Kabir, P.M. Bhatt, R.K. Malhan, P. Rajendran, B.C. Shah, and S.K. Gupta. Task Assignment and Motion Planning for Bi Manual Mobile Manipulation. IEEE International Conference on Automation Science and Engineering (CASE), Vancouver, Canada, August 2019.

[19] R.K. Malhan, A.M. Kabir, A.V. Shembekar, B. Shah, T. Centea, and S.K. Gupta. Hybrid Cells for Multi-Layer Prepreg Composite Sheet Layup. IEEE International Conference on Automation Science and Engineering (CASE), Munich, Germany, August 2018.

[20] R.K. Malhan, Y. Shahapurkar, A.M. Kabir, B. Shah, and S.K. Gupta. Integrating impedance control and learning based search scheme for robotic assemblies under uncertainty. ASMEs 13th Manufacturing Science and Engineering Conference, College Station, Texas, USA, June 2018.

[21] R.K. Malhan, A.M. Kabir, B. Shah, T. Centea, and S.K. Gupta. Automated Prepreg Sheet Placement Using Collaborative Robotics. Society for the Advancement of Material and Process Engineering (SAMPE) Technical Conference and Exhibition, Long Beach, CA, May 2018.

[22] A.M. Kabir, A.V. Shembekar, R.K. Malhan, R.S. Aggarwal, J.D. Langsfeld, B.C. Shah, and S.K. Gupta. Robotic finishing of interior regions of geometrically complex parts. ASMEs 13th Manufacturing Science and Engineering Conference, College Station, Texas, USA, June 2018.